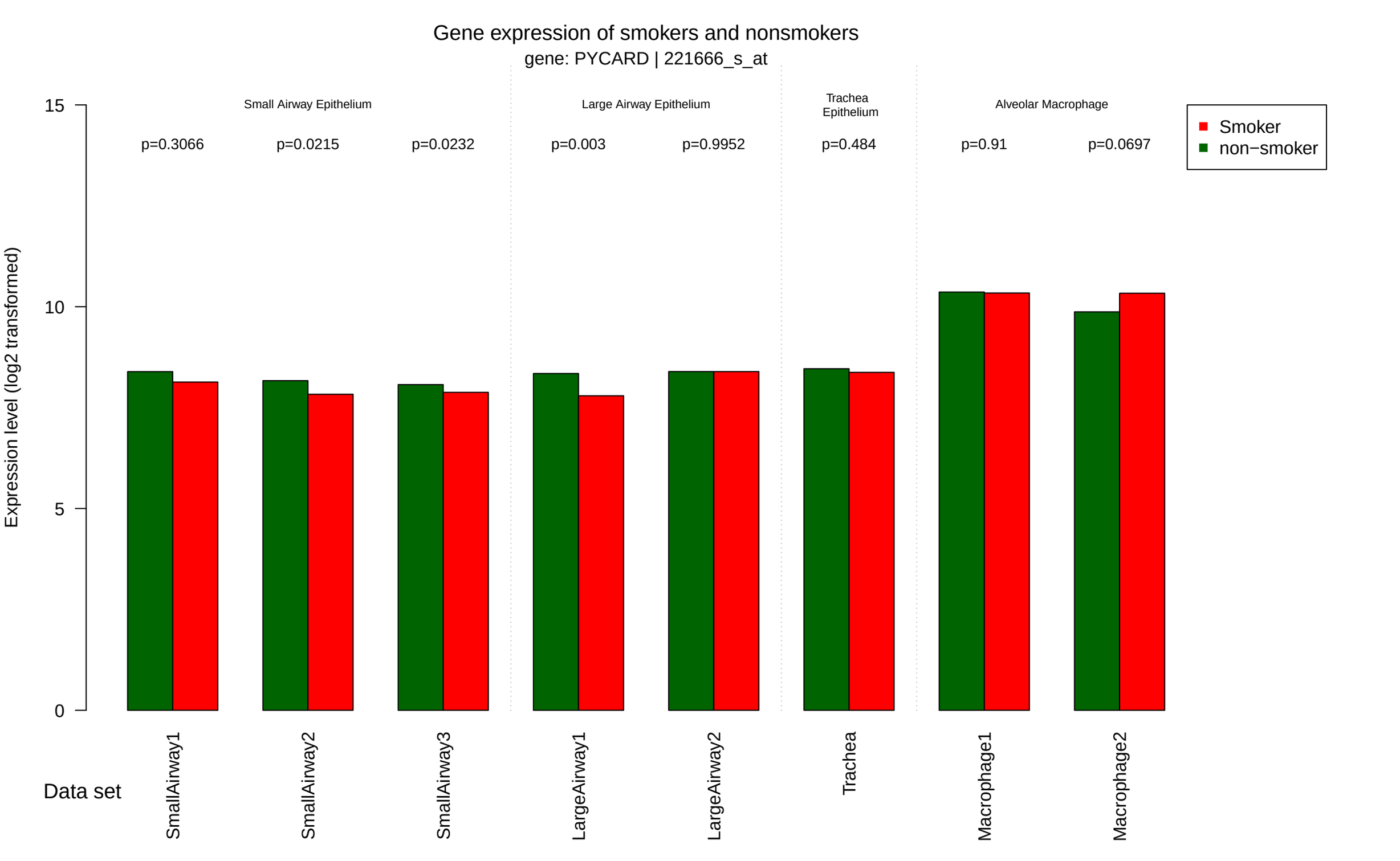
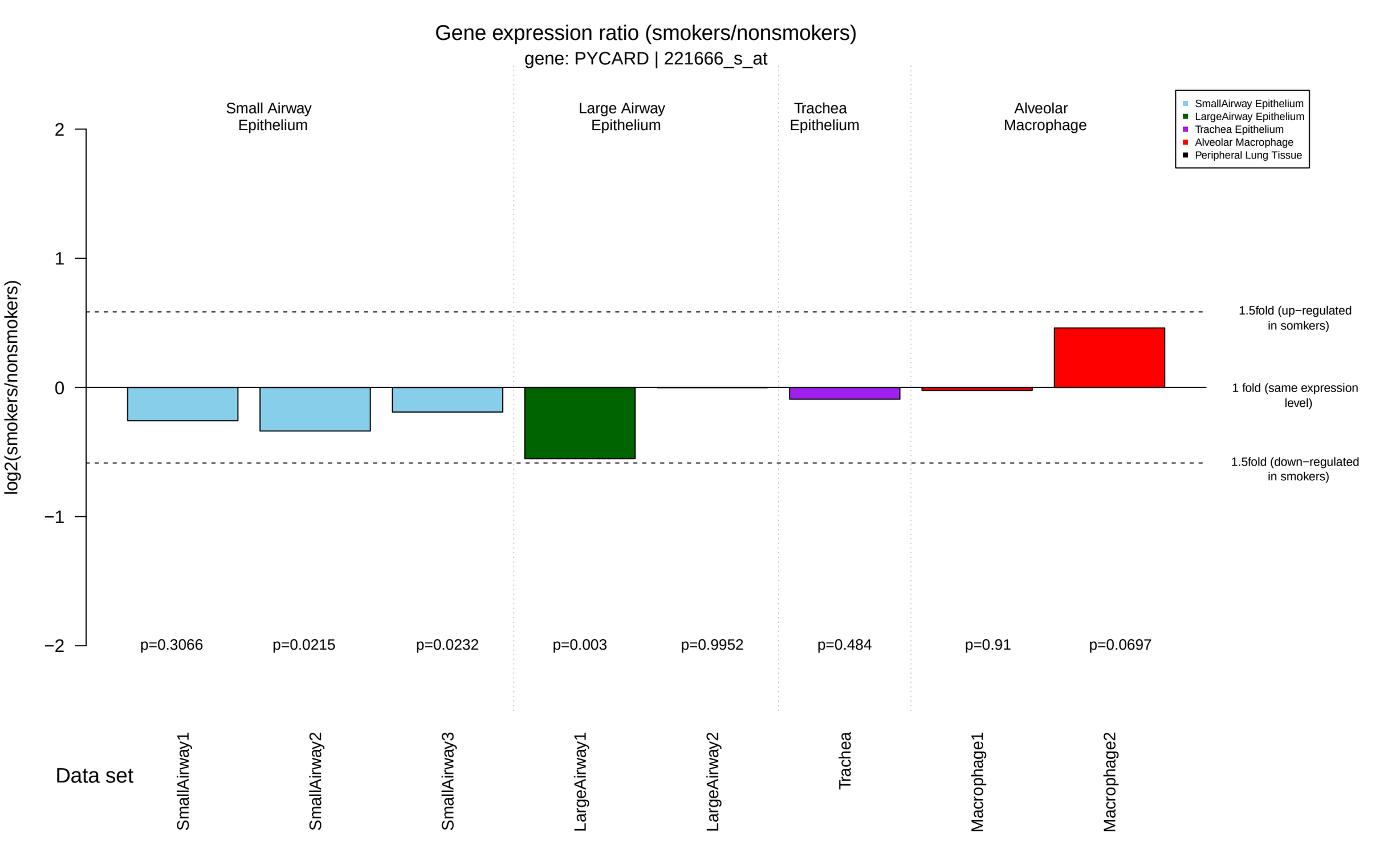
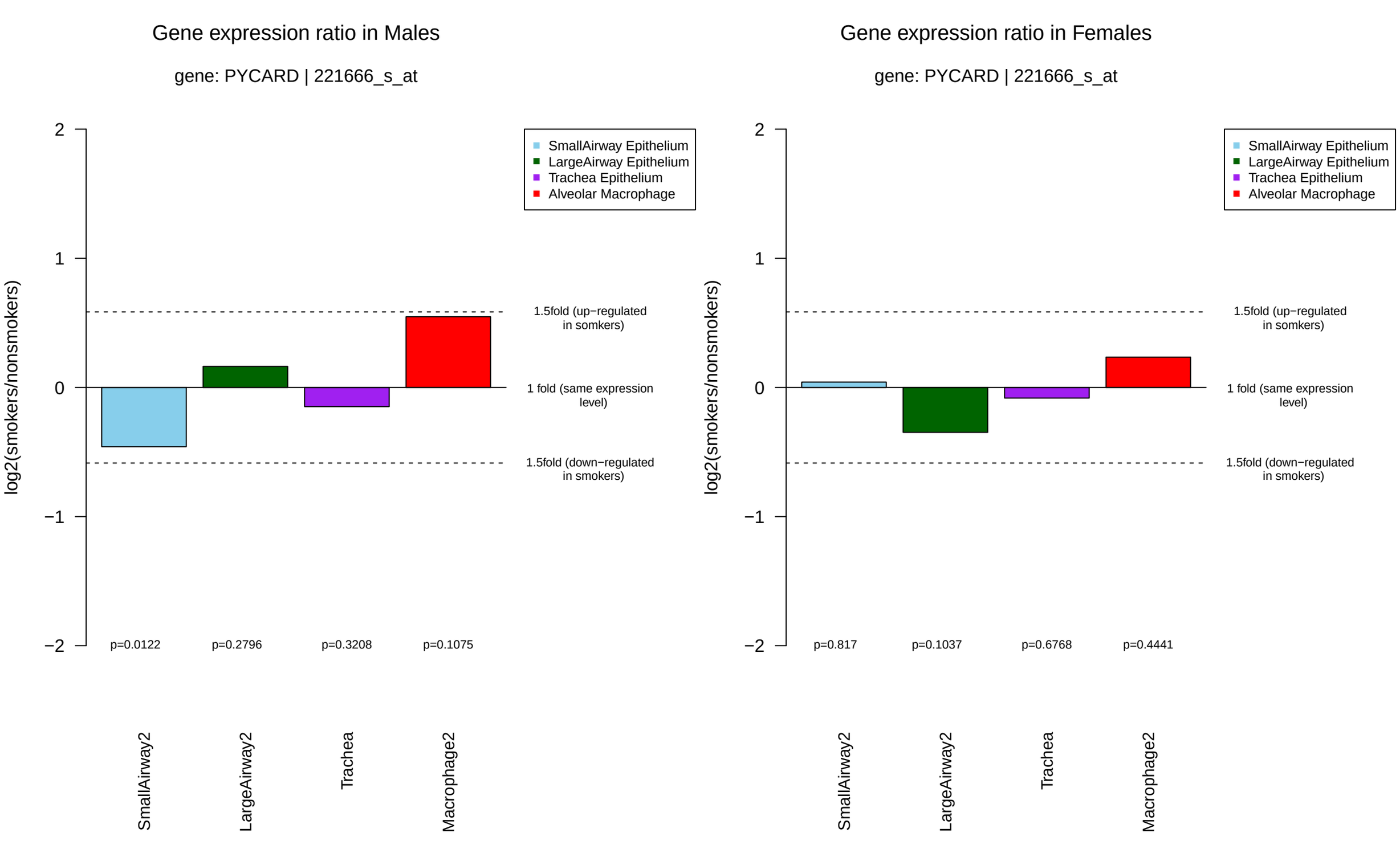
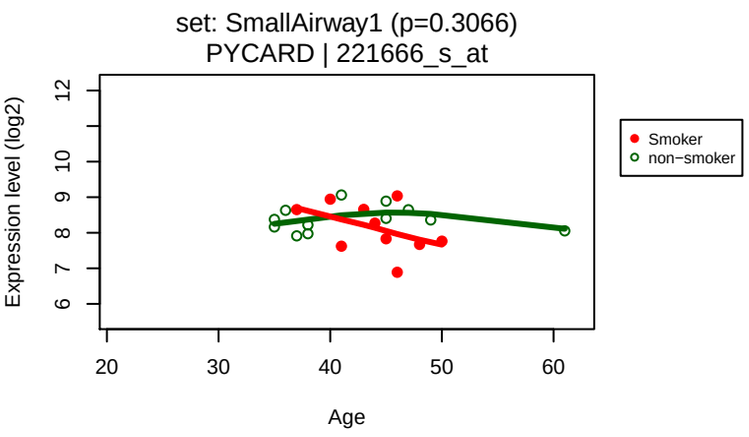
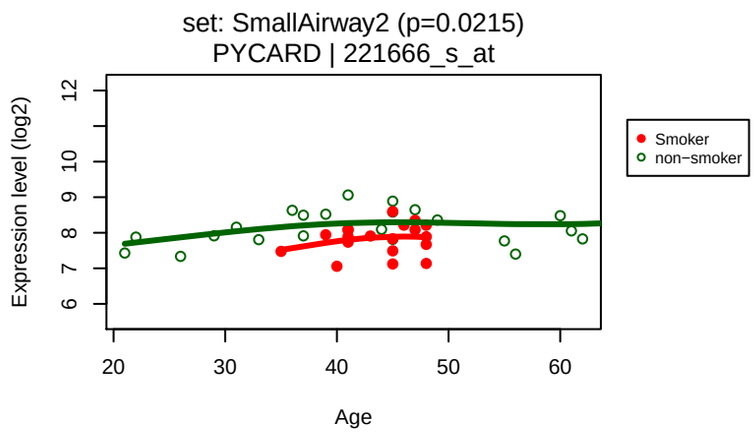
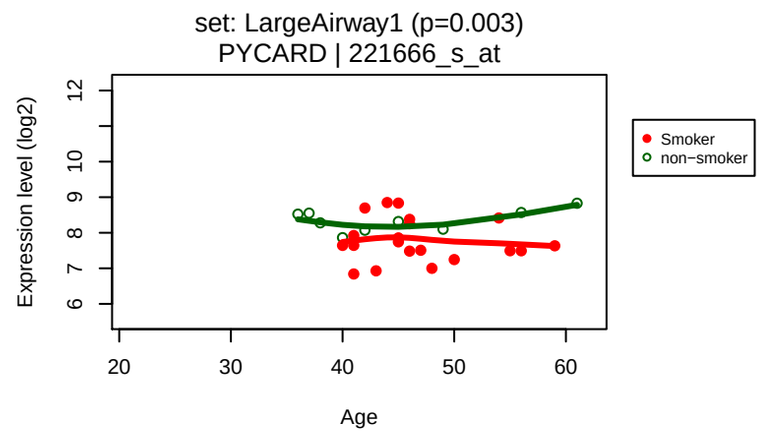

Gene expression of smokers and nonsmokers
gene: PYCARD | 221666_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3066
p=0.0215
p=0.0232
p=0.003
p=0.9952
p=0.484
p=0.91
p=0.0697
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

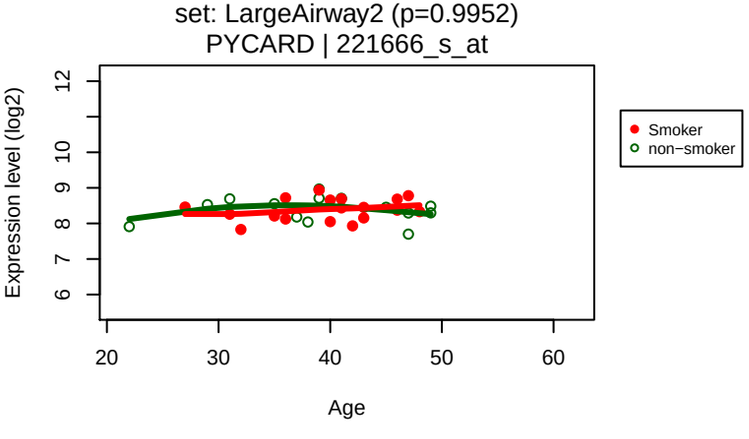

Gene expression ratio (smokers/nonsmokers)
gene: PYCARD | 221666_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3066
p=0.0215
p=0.0232
p=0.003
p=0.9952
p=0.484
p=0.91
p=0.0697
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

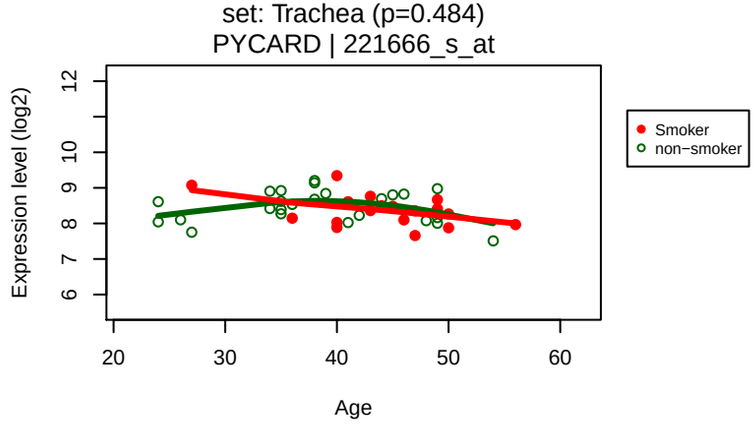

Gene expression ratio in Males
gene: PYCARD | 221666_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0122
p=0.2796
p=0.3208
p=0.1075
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PYCARD | 221666_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.817
p=0.1037
p=0.6768
p=0.4441
SmallAirway2
LargeAirway2
Trachea
Macrophage2

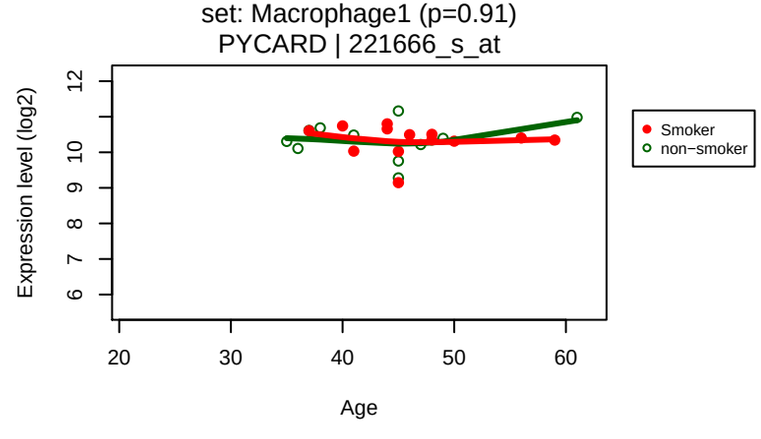

set: SmallAirway1 (p=0.3066)
PYCARD | 221666_s_at
Expression level (log2)
Age
Smoker
non−smoker

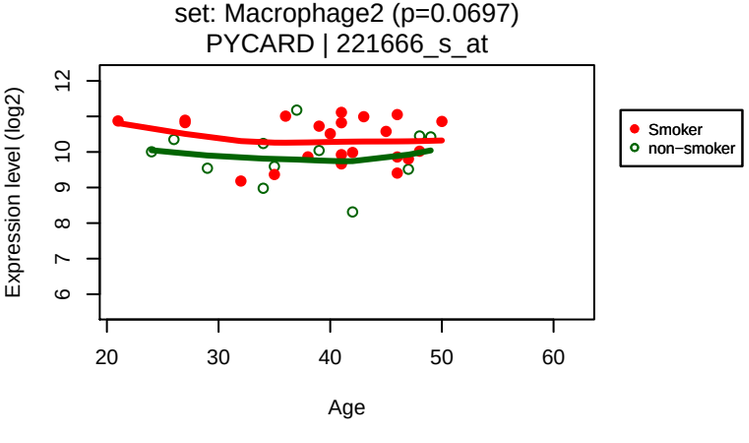

set: SmallAirway2 (p=0.0215)
PYCARD | 221666_s_at
Expression level (log2)
Age
Smoker
non−smoker

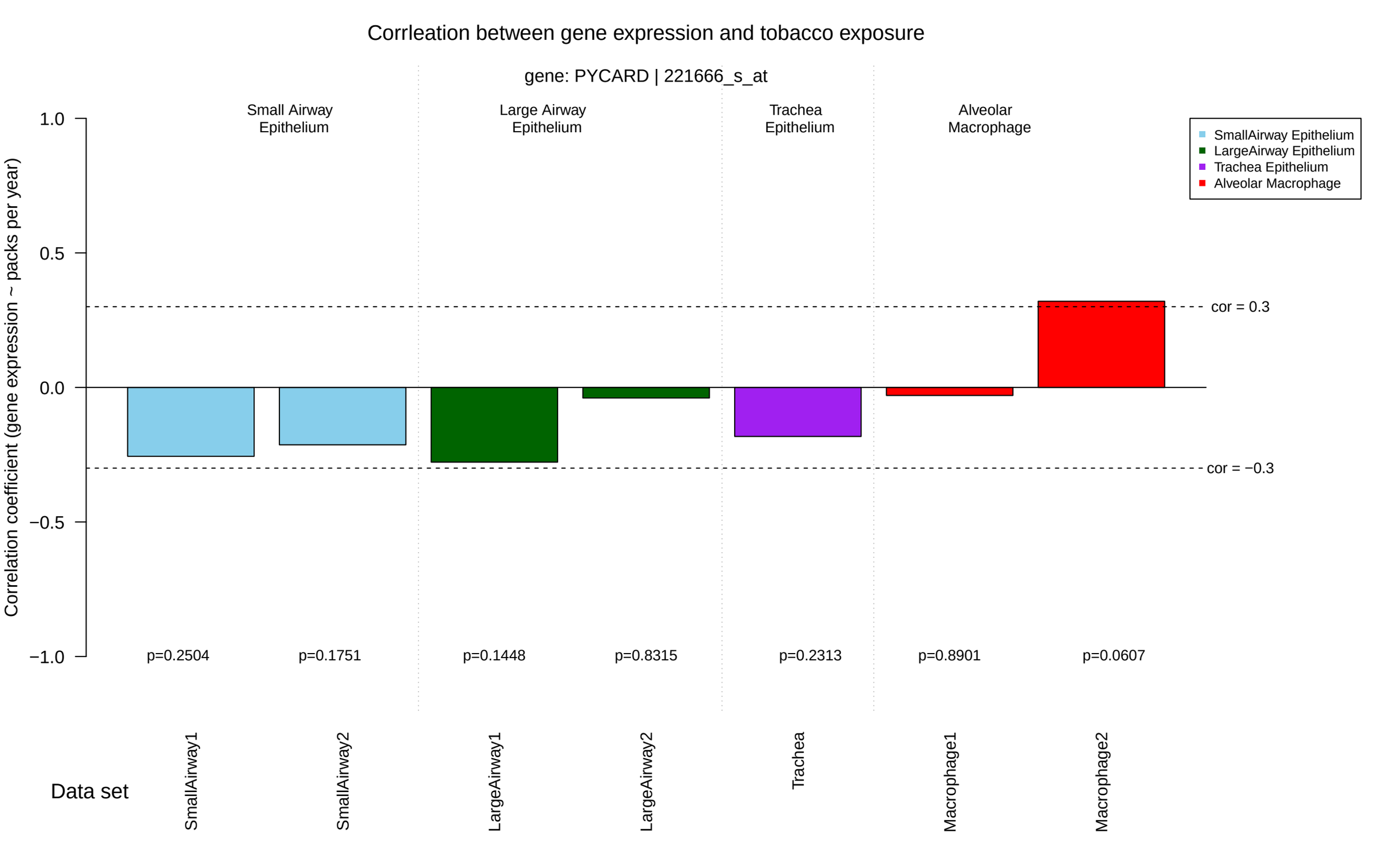

set: LargeAirway1 (p=0.003)
PYCARD | 221666_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9952)
PYCARD | 221666_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.484)
PYCARD | 221666_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.91)
PYCARD | 221666_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0697)
PYCARD | 221666_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PYCARD | 221666_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2504
p=0.1751
p=0.1448
p=0.8315
p=0.2313
p=0.8901
p=0.0607
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2